Data & AI Builders Conference

Building modern data foundations for intelligent solutions together

AI where business data lives

In SQL Server 2025 and Azure SQL

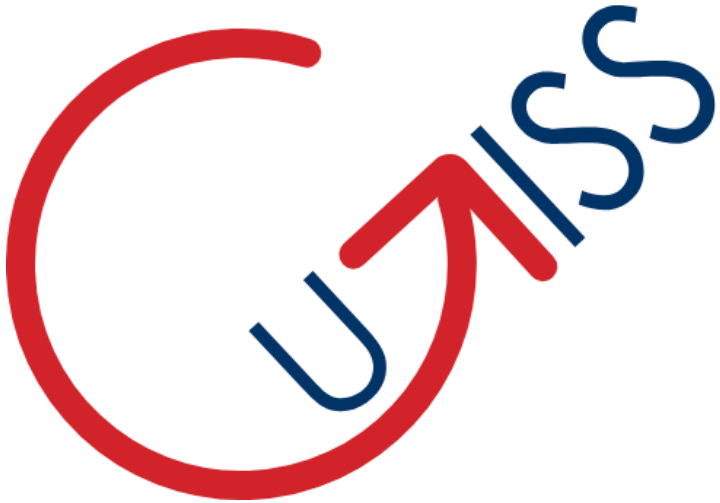
Slide and demo: bit.ly/dab-2026-sgovoni



Sponsors



Collaborators







Microsoft
Most Valuable Professional

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Lodestar

Agenda



- Introduction to AI-ready evolution of SQL Server 2025 and Azure SQL
- Using vector data types and perform semantic search
  - Preparing SQL instance
  - Preparing data for semantic search
  - Generating embeddings with Azure OpenAI and T-SQL
  - Perform semantic search with VECTOR_DISTANCE, VECTOR_SEARCH
- Vector indexes in Azure SQL Database



01

Introduction to AI-ready evolution



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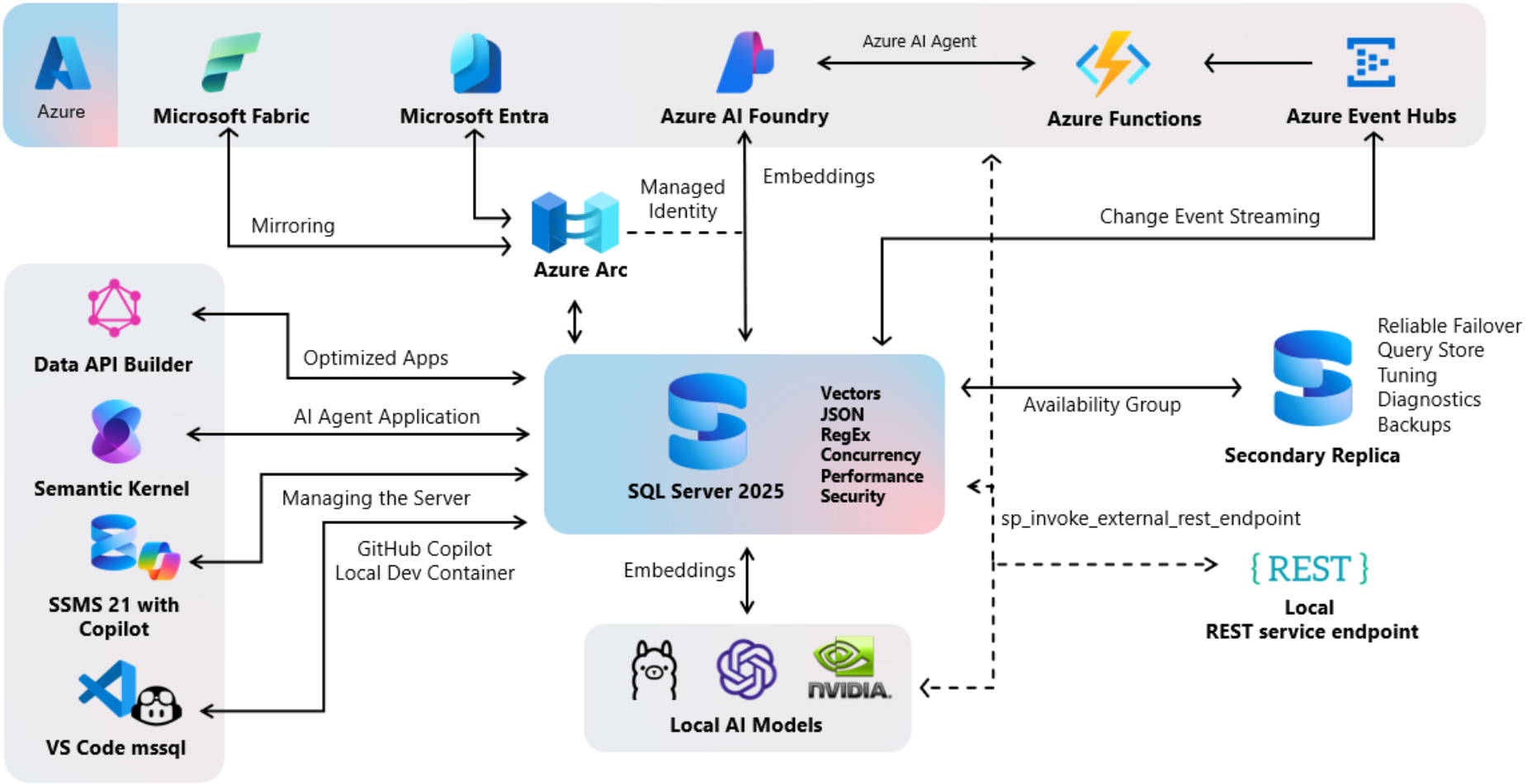
SQL Server 2025 AI capabilities



- SQL Server 2025 introduces AI capabilities directly into the database engine
  - Vector data type
  - Connectivity with Azure OpenAI and other frameworks
  - Embedding generation
  - Vector search and vector indexes
  - sys.sp_invoke_external_rest_endpoint (available from SQL Server 2022)
  - Foundry models sold by Azure



SQL Server 2025 AI capabilities





02

Vector data types and semantic search

From zero to AI

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Preparing SQL instance



- The sys.sp_invoke_external_rest_endpoint system stored procedure is available in SQL Server 2025 and Azure SQL Managed Instance, but it's disabled by default
  - It's enabled by default in Azure SQL Database and in SQL Fabric

- To call external REST endpoint, you must enable the external rest endpoint enabled server configuration option using sp_configure



Preparing data for semantic search

- Raw data is usually optimized for relational queries, filters, and joins
- Semantic search needs a richer text built from relevant content and metadata
- A dedicated table helps prepare and control the text used to generate embeddings
- This improves search quality by making data searchable by meaning, not only by keywords



Generating embeddings with Azure OpenAI and T-SQL



- Use an EXTERNAL MODEL to connect SQL Server to the embedding model endpoint
- Generate embeddings directly from T-SQL with AI_GENERATE_EMBEDDINGS
- Store the vector representation next to the prepared text in the database
- Process data in batches to control execution time, throughput, and cost



Introduction to semantic search

- Text can be transformed into embeddings, which are vectors that represent meaning
- Similar concepts produce vectors that are close to each other
- A natural language question is converted into an embedding that represents its meaning
- The embedding of the question is compared with stored embeddings to find semantically similar content



Introduction to semantic search

- Semantic search can be combined with traditional SQL filters and metadata
- Results are ranked by distance: lower distance means higher semantic similarity
- Execution plans are key to understanding how semantic search queries behave
- Performance depends on whether the query uses exact distance calculation or indexed approximate search



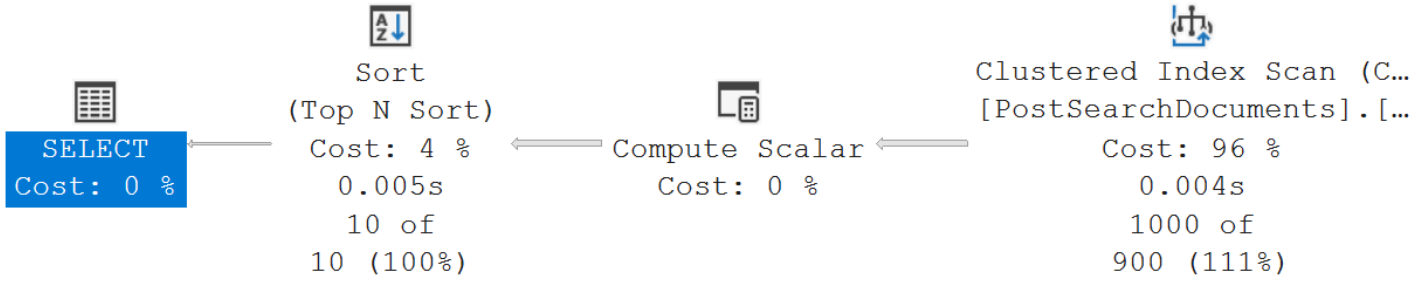
Semantic search with VECTOR_DISTANCE



- VECTOR_DISTANCE is designed for exact similarity checks, not approximate retrieval
- Every candidate embedding must be evaluated against the embedding of the “question”
- Existing ANN vector indexes do not change this behavior
- Best suited for scenarios where exact results matter and the candidate set is controlled



Semantic search with VECTOR_DISTANCE



- The query optimizer scans the candidate rows instead of performing an index seek
- VECTOR_DISTANCE is evaluated for each candidate embedding
- A Compute Scalar operator performs the distance calculation
- Sorting is needed to return the closest matches first
- This behavior is transparent and predictable, but less scalable on large candidate sets

Vector indexes

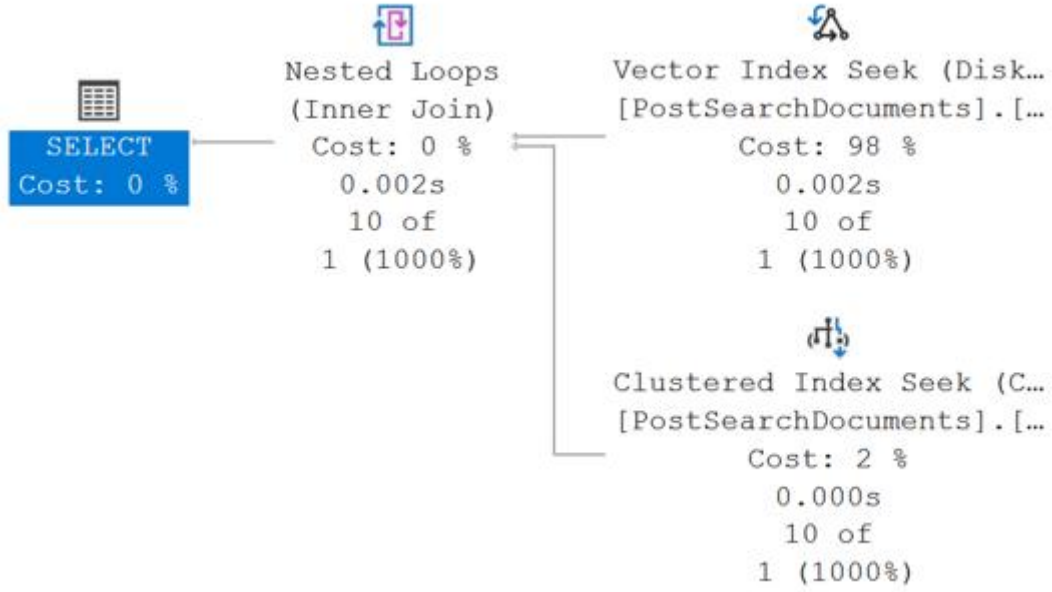


- A vector index
  - Helps SQL Server search for similar embeddings more efficiently
  - Is built to support ANN (approximate nearest-neighbor) search
- SQL Server uses an algorithm known as DiskANN to navigate similar embeddings more efficiently
- The goal is to reduce the cost of scanning large candidate sets
- Performance improves by accepting a controlled approximation



Semantic Search with VECTOR_SEARCH

- To use the vector index, the query must follow specific requirements
  - The use of VECTOR_SEARCH
  - VECTOR_SEARCH must be used with TOP (N) WITH APPROXIMATE
  - Results must be ordered by the distance column
- When the requirements are met, the execution plan can use vector index seek
- Without approximate search, the optimizer can fall back to a scan



Vector index seek in the execution plan

Important considerations and limitations



- Vector index features are still in preview in SQL Server 2025
- In SQL Server 2025, creating a vector index currently makes the table read-only
  - INSERT, UPDATE, DELETE, and MERGE operations are not supported on that table after index creation
- Relational filters may be applied after vector search, depending on the index version



Important considerations and limitations



- In vector search, SQL Server may fetch more vectors than requested before applying final ranking
  - This over-fetching helps reduce the approximation error
- Always validate syntax, limitations, and execution plans in your environment





03

Vector indexes in Azure SQL DB

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Vector Indexes: SQL Server 2025 vs Azure SQL



- SQL Server 2025 currently uses the first vector index implementation
- Azure SQL Database uses the latest vector index format, version 3
- The index version can be checked through sys.vector_indexes
- The same CREATE VECTOR INDEX syntax can lead to different behaviors depending on the platform



Vector Indexes: DML Support



- In SQL Server 2025, creating a vector index currently makes the table read-only
  - INSERT, UPDATE, DELETE, and MERGE fail after the vector index is created
- In Azure SQL Database, vector-indexed tables remain read-write
  - Azure SQL Database supports DML with automatic vector index maintenance
  - Azure SQL Database also supports TOP (N) WITH APPROXIMATE and iterative relational filtering



04

Demo



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Summary



- Embeddings stored close to relational data enable semantic search directly with T-SQL
- VECTOR_DISTANCE performs exact search: every candidate is evaluated
- VECTOR_SEARCH with a vector index performs approximate nearest-neighbor search
- SQL Server 2025 still has preview limitations, including read-only tables after CREATE VECTOR INDEX
- Azure SQL Database uses vector index version 3, keeps tables read-write, and supports iterative filtering



Resources



- VECTOR_DISTANCE
- VECTOR_SEARCH
- AI where business data lives
  - Part 1: Preparing Data for Semantic Search
  - Part 2: Generating Embeddings with Azure OpenAI and T-SQL
  - Part 3: Semantic Search with T-SQL
  - Part 4: Indexed Vector Search with SQL Server 2025
  - Part 5: VECTOR_DISTANCE vs VECTOR_SEARCH: How Does the Execution Plan Change? (thanks to Riccardo Perico)
  - Part 6: coming soon ☺



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THANK YOU !!



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SQL Server 2025 e Azure SQL: l'AI arriva dove vivono i dati aziendali